

SAIKUMAR KALERU

C++ / LOW LATENCY SYSTEMS ENGINEER

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PROFESSIONAL SUMMARY

C++ systems engineer with a B.E. in Computer Science and a solid grip on data structures, algorithms, OOP, and operating systems. I work in modern C++20 with a focus on multithreading, concurrency, and memory and performance tuning. Spent close to two years at D. E. Shaw as Member Technical on a Futures and Options trading desk, building C++ and Python systems for a real time environment. I wrote a C++20 limit order book with a price and time priority matching engine, and I profile and benchmark on Linux with CMake. Ranked in the top 5% in GATE CSE 2021.

TECHNICAL SKILLS

Languages: C++ (C++20), Python, SQL, Java, Bash

Modern C++: RAII, templates, move semantics, STL, smart pointers

Concurrency & Low Latency: multithreading, synchronization, lock free design, cache aware data layout, memory and performance tuning

Build & Tools: CMake, profiling, benchmarking, pytest, Git, WebSockets

Platform & Data: Linux, Docker, SQL, PostgreSQL, MS SQL

Core CS: Data Structures & Algorithms, OOP, Operating Systems, Computer Networks, System Design

EXPERIENCE

D. E. Shaw India Pvt. Ltd. | Member Technical

May 2022 to Jan 2024

C++ and Systems Engineering, F&O Trading Desk

- Built and maintained C++ and Python systems for a Futures and Options trading desk.
- Wrote automated data pipelines to capture and process live market data.
- Used multithreading and profiling to keep latency sensitive code fast and stable.
- Built analytics tools and daily reports that the desk used to track the market.

Arcesium India Pvt. Ltd. | Site Reliability Engineer (Intern)

Jan 2022 to May 2022

Backend Reliability and Databases

- Kept AWS EC2 and RDS services available and fixed production database issues.
- Managed MS SQL and PostgreSQL databases, improved slow queries, and tested Flyway for schema migration.

PROJECTS

Low Latency Order Book | C++20, Python, WebSockets

- Built a limit order book with a price and time priority matching engine in C++20, using cache aware data structures and tight hot paths for insert, cancel, and match.
- Added Python bindings and a WebSocket server that streams order book state to a web GUI.

REST API Service | Python, FastAPI, Docker

- Built a REST API with token based login, a relational database, and pagination.
- Wrote a pytest test suite and packaged the service with Docker.

Ecommerce Microservices | Docker, Kubernetes, Jenkins

- Built an ecommerce backend as containerized microservices on Kubernetes, with a Jenkins pipeline to build, test, and deploy them.

EDUCATION

B.E. Computer Science & Engineering, CBIT, Hyderabad

2018 to 2022

Chaitanya Bharathi Institute of Technology · CGPA 8.8

MBA in Finance, UCCBM, Hyderabad

2024 to 2026

University College of Commerce & Business Management · CGPA 8.9

RELEVANT COURSEWORK

Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, Object Oriented Programming, Design & Analysis of Algorithms, Software Engineering, Computer Architecture

CERTIFICATIONS

HackerRank: Problem Solving and Java (Basic) Programming

NISM Series-XV: Research Analyst (SEBI/NISM), Reg. No. NISM-202600123995

ACHIEVEMENTS

- GATE CSE 2021: top 5% among national candidates in Computer Science.
- UGC NET 2025 (Management): 95.6 percentile.

ADDITIONAL INFORMATION

Languages: English, Hindi, Telugu · Location: Hyderabad, India (open to relocation) · Availability: Immediate